

Timing Matters: Step-Synchronized Mastoid Vibration Improves Turning Detectability in Redirected Walking

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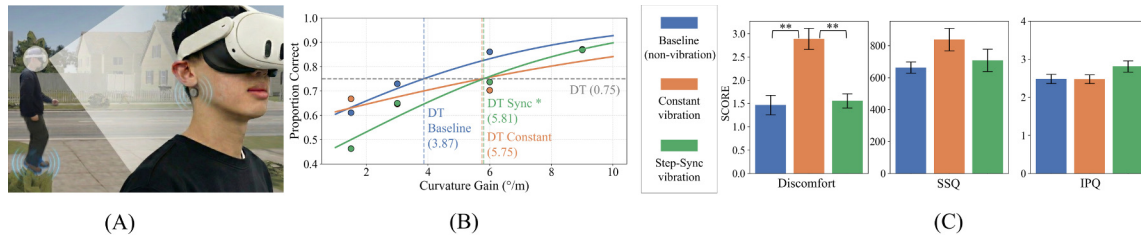


Figure 1: (A) Step-synchronized mastoid vibration delivered at each detected footstep. (B) Psychometric detection curves showing 75% detection thresholds (DT) for Baseline, Constant, and Sync conditions; dots indicate observed proportion-correct responses. (C) Subjective ratings of Discomfort, SSQ, and IPQ with SD error bars (* $p < .05$, ** $p < .01$).

ABSTRACT

Redirected walking is limited by sensitivity to unintended turning. Mastoid vibration reduces detectability, but constant stimulation increases discomfort. We examined if step-synchronized vibration improves tolerance without extra discomfort. Participants performed tasks under baseline, constant, and step-synchronized conditions. Step-synchronized vibration increased detection thresholds by 50% over baseline with similar discomfort. Constant vibration showed comparable benefits but higher discomfort. Neither condition affected presence or simulator sickness. These results suggest that step-synchronized vibration shows potential to expand redirected walking while preserving user comfort.

Index Terms: Redirected walking; Mastoid vibration; Gait-synchronized feedback; Detection threshold; Psychophysics; Virtual reality locomotion

1 INTRODUCTION

Natural walking is central to presence in virtual reality (VR), yet most systems operate within spaces far smaller than the environments they simulate. Redirected walking (RDW) addresses this constraint by subtly curving the user's physical trajectory while preserving the intended virtual path. The effectiveness of RDW is limited by turning perception, commonly expressed as detection thresholds (DT) for curvature gains. Higher DT values allow stronger redirection without users noticing, improving spatial efficiency without compromising experience.

Recent study suggests that vestibular-adjacent mastoid vibration can increase curvature DTs, making redirection more difficult to detect [1]. However, always-on stimulation has been associated with increased wear-related discomfort [2]. In contrast, research on cybersickness has shown that step-synchronized bilateral mastoid vibration can improve comfort compared to continuous stimulation [2], suggesting that gait-synchronized mastoid vibration may reduce vibration-related discomfort in RDW without necessarily compromising its effects on turning perception.

Building on this observation, we investigate how the temporal alignment of bilateral mastoid vibration influences the perception of

unintended turning during active walking. By comparing constant and step-synchronized stimulation conditions against a no-vibration baseline, this study examines whether gait-coupled timing can meaningfully affect users' sensitivity to curvature gains while mitigating the discomfort often associated with continuous vibration. Through this investigation, we aim to determine whether the timing of mastoid stimulation can serve as a practical parameter for balancing redirection tolerance and user comfort in redirected walking.

2 RELATED WORK

Prior work formalized curvature gains and established DT as a practical upper bound on how much redirection can remain unnoticed during active walking [3]. Subsequent surveys further examined factors influencing these thresholds and emphasized perceptually grounded gain scheduling [4].

On the vestibular side, mastoid bone-conduction vibration has been proposed as a method capable of elevating DTs while preserving presence [1]. However, continuous vibration may exacerbate simulator sickness and wearing discomfort.

Outside the RDW domain, research on locomotion and cybersickness has shown that aligning vibration timing with users' gait—particularly through bilateral step-synchronized patterns—can enhance comfort and perceptual stability [2]. Despite these findings, the temporal organization of mastoid vibration has not been systematically investigated within RDW, where perceptual thresholds and comfort dynamics may differ.

This gap motivates the present study, which investigates whether the temporal alignment of mastoid vibration can meaningfully influence curvature-gain detectability and user comfort in RDW.

3 METHODS

3.1 Experimental Setup

Experiments were conducted along a straight, unobstructed 5 m walkway. Visual stimuli were rendered on a Meta Quest 3 HMD, spatially aligned with the physical path. The virtual environment was developed in Unity 6000.0.34f1 and presented through the headset.

Bilateral vibrotactile actuators were attached to the mastoid region using a lightweight headband and driven at 150 Hz—a frequency previously reported to be non-discomforting for head-worn mastoid stimulation [5]. A total of nine participants (3 female, 6 male; age = 22–28 years, $M = 24.9$, $SD = 2.1$) took part in the study.

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3.2 Experimental Design

We employed a within-subjects $3(\text{Vibration}) \times 5(\text{Curvature Gain})$ factorial design. The *Vibration* factor included three levels: (i) *Baseline* (no vibration), (ii) *Constant* (always-on), and (iii) *Sync* (step-synchronized).

Curvature gain was applied following standard RDW procedures as a continuous rotation of the virtual environment during forward walking and was expressed in degrees per meter walked ($^{\circ}/\text{m}$) [3]. The *Curvature Gain* factor comprised five levels: 0, ± 1.5 , ± 3 , ± 6 , and $\pm 9^{\circ}/\text{m}$, where the sign indicated turning direction.

Each condition was repeated six times (three per direction), yielding $3 \times 5 \times 6 = 90$ trials in total. Vibration conditions were block-randomized using a Latin-square counterbalancing scheme where applicable, with short breaks provided between blocks. At the start of each block, vibration amplitude was individually calibrated to the highest intensity below self-reported discomfort.

For the step-synchronized condition, a brief per-subject calibration was conducted in which each participant walked straight for approximately 5 s while the system learned a personalized step template from head-mounted IMU vertical acceleration. During trials, short bilateral vibration bursts were triggered whenever an online step detector matched this template, with a refractory period to prevent double triggering.

3.3 Task

At the start of each trial, participants stood at a designated start position and were instructed to walk toward a red target sphere located 5 m straight ahead. Upon reaching the end line, participants reported their perceived turning direction using a two-alternative forced-choice (2AFC) method (left vs. right), prioritizing accuracy. After each vibration block, participants completed the Simulator Sickness Questionnaire (SSQ), a wearing-related discomfort rating, and the Igroup Presence Questionnaire (IPQ) subscales—Spatial Presence, Involvement, and Realism [6, 7]. Subscale scores were averaged to form a composite presence measure for statistical analysis.

3.4 Data Analysis

For each participant and vibration condition, 2AFC responses across curvature gains were fitted with a cumulative logistic psychometric function using maximum likelihood estimation. The 75% DT, commonly used as a criterion for reliable discrimination, was extracted as the primary measure of turning sensitivity [3].

DT values were analyzed using a two-way repeated-measures ANOVA with factors *Vibration* (Baseline, Constant, Sync) and *Curvature Gain*. When the assumption of sphericity was violated, Greenhouse–Geisser corrections were applied, and Bonferroni-adjusted post-hoc tests were used for pairwise comparisons. Subjective measures (SSQ total score, wearing discomfort rating, and composite IPQ presence score) were each analyzed using a one-way repeated-measures ANOVA with factor *Vibration*. The significance level was set at $\alpha = 0.05$.

4 RESULTS

A two-way repeated-measures ANOVA was conducted to investigate the effects of *Vibration* (Baseline, Constant, Sync) and *Curvature Gain* on DT. The analysis revealed significant main effects of *Vibration*, $F(2, 16) = 5.18$, $p = .018$, $\eta_p^2 = .39$, and *Curvature Gain*, $F(3, 24) = 12.93$, $p < .001$, $\eta_p^2 = .62$. In contrast, the *Vibration* $\times$ *Gain* interaction was not significant, $F(6, 48) = 1.37$, $p = .25$.

Bonferroni-adjusted post-hoc comparisons were performed to further evaluate differences among vibration conditions. The *Sync* condition produced significantly higher DTs than *Baseline* ($p = .024$), whereas all other pairwise comparisons were not significant (all $p > .29$; see Fig. 1(B)).

For subjective measures, SSQ total scores did not differ significantly across vibration conditions, although a trend was observed

($p = .087$). In contrast, wearing discomfort showed a strong main effect of *Vibration*, $F(2, 16) = 26.94$, $p < .001$, $\eta_p^2 = .77$. Post-hoc tests indicated that constant vibration induced higher discomfort than both *Baseline* and *Sync* (all $p < .001$), while no difference was found between *Baseline* and *Sync* ($p = .40$).

Finally, presence scores showed no significant main effect of *Vibration*, $p = .92$.

5 CONCLUSION

This pilot study indicates that mastoid vibration during RDW has the potential to reduce turning detectability, with both constant and step-synchronized patterns showing comparable effects on turning DT. However, the two vibration strategies differed in their subjective consequences. While step-synchronized stimulation was associated with increased turning thresholds without a corresponding rise in discomfort, constant stimulation tended to induce higher discomfort despite similar performance gains. These observations suggest that perceptual modulation through mastoid vibration may be achieved without continuous stimulation, and that temporal alignment with gait events could represent a more efficient and user-tolerant approach.

Importantly, presence and simulator sickness did not exhibit notable degradation relative to baseline, implying that the observed improvements in turning detectability can be attained without substantially compromising user experience.

Further investigation with larger and more balanced participant samples, as well as refined validation of the gait-synchronized stimulation pipeline, will be necessary to more firmly establish the robustness and generalizability of the observed effects.

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